

Geometric areas



1

a 12 units²

b 21 units²

c 4 units²

d 12 units²

2

a 18

b 72

c 104

d 38.5

e 18

f 80

g 18 π

h 9 π

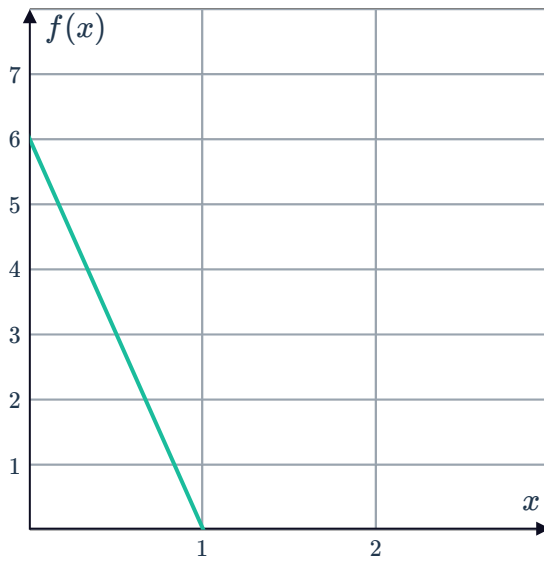
i 32 π

j 18 π

3

a

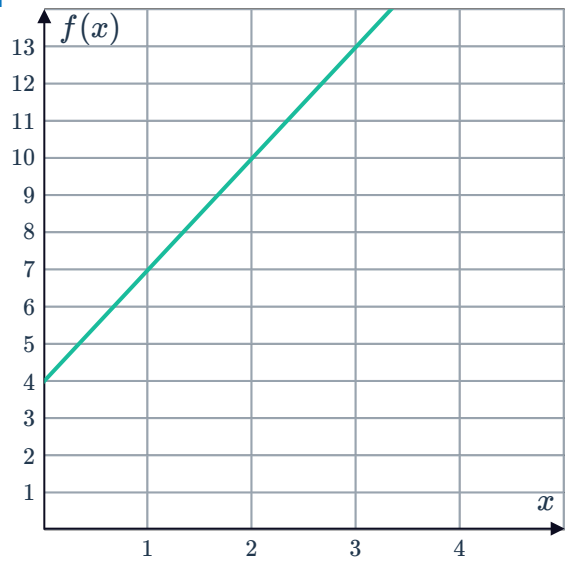
i



ii 3 units²

b

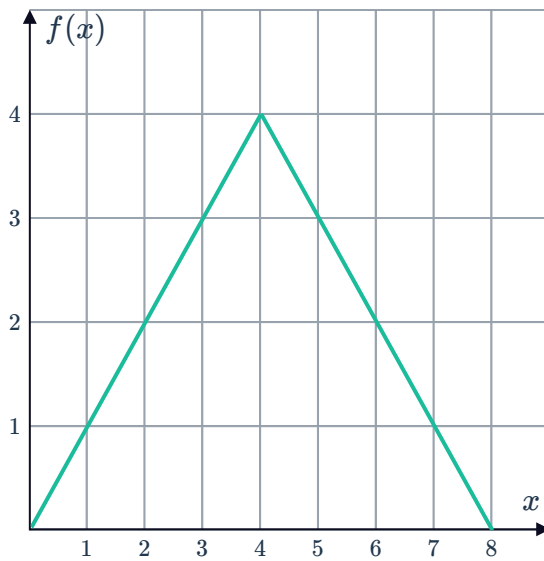
i



ii $\frac{51}{2}$ units²

c

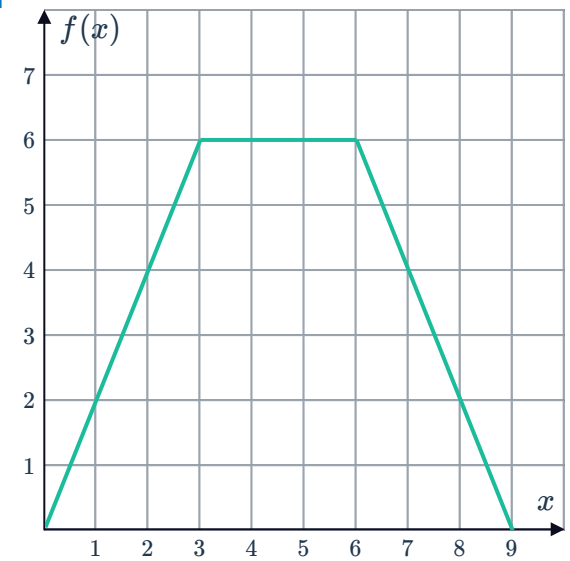
i



ii 16 units²

d

i



ii 36 units²

Approximate areas



4

- a An overestimation as the function is increasing on the given interval.
- b An underestimation as the function is decreasing on the given interval.

5

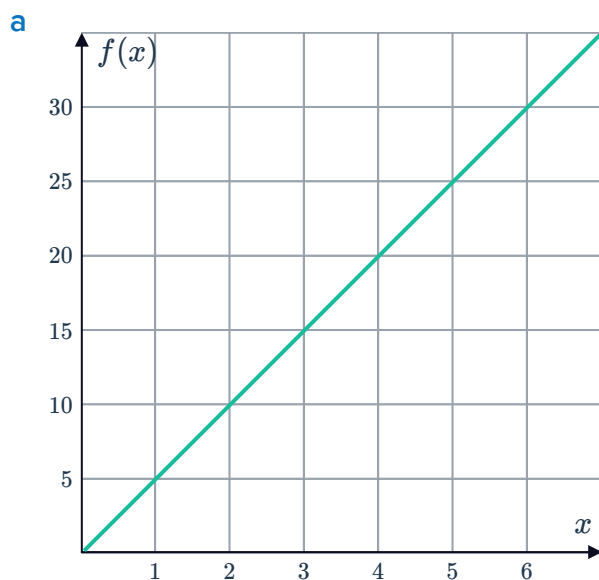
- a An underestimation as the function is increasing on the given interval.
- b An overestimation as the function is decreasing on the given interval.

6 The left endpoint approximation because the function is always increasing.

7

- a
 - i Overestimate
 - ii Underestimate
 - iii Underestimate
 - iv Overestimate
- b
 - i Underestimate
 - ii Overestimate
 - iii Overestimate
 - iv Underestimate

8

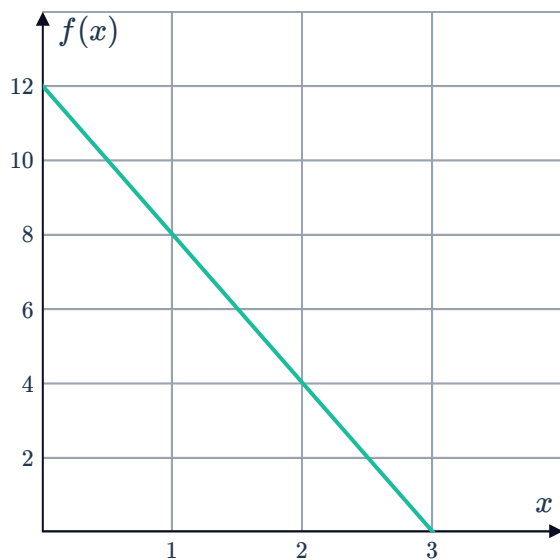


- b
- i 60 units²
 - ii 120 units²

- c
- i 75 units²
 - ii 105 units²
- d $A = 90 \text{ units}^2$

9

a



b

i 24 units²ii 12 units²

c

i 21 units²ii 15 units²d 18 units²

10

a 62 units²b 50 units²11 336 units²12 $\frac{77}{60}$ units²

13

a 432 units²b 576 units²

Use technology

14

a 30

b $e^3 - 1$

c 36

15

a

n	5	10	100	1000	10 000
A_L units ²	3.560	3.940	4.293	4.329	4.333

b $\frac{13}{3}$ units²

16



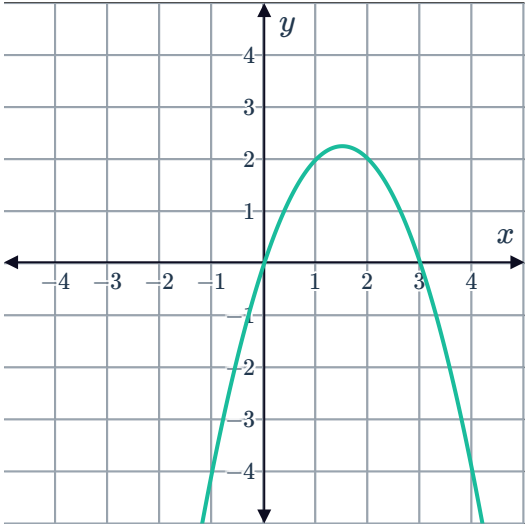
a

n	5	10	100	1000
$A_R \text{ units}^2$	2.63705	2.90452	3.12042	3.13956

b $\pi \text{ units}^2$

17

a



b $0 \leq x \leq 3$

c

n	5	10	100	1000	10 000
$A_R \text{ units}^2$	4.3200	4.4550	4.4996	4.5000	4.5000

d $\frac{9}{2} \text{ units}^2$